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SHORT PROLINE-SUBSTITUTED MUC1 MUCIN PEPTIDES CAN BIND TO MOUSE MUC1 MONOCLONAL ANTIBODY AS REVEALED BY STD NMR

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Abstract

Many studies on the biological activity of mucin peptides derived from the VNTR domain that are considered as potential cancer vaccines have involved and focused on the immunodominant epitope PDTRP. A short peptide GVTSAPD preceding the main binding epitope and six other substituted peptides were investigated for their ability to bind mouse Muc1 monoclonal antibody (mAb, 6A4) by STD NMR. The Pro6 residue of GVTSAPD was replaced by Asp, Leu, Tyr, Phe and Trp to obtain the pro-substituted peptides. A sixth mutant peptide with Ser4 replaced by Asn was also studied for comparison. The results showed that substitution of Ser4 by Asn showed similar binding properties to the native peptide. The pro-substituted peptides with position-6 occupied by Phe or Trp can bind the mAb, but not those containing Asp or Leu. The pro-substituted peptides GVTSAFD and GVTSAWD may potentially be useful as additional epitopes toward the development of cancer vaccine.

Keywords: Muc1 antibody recognition epitope, Mucin peptide, Saturation Transfer Difference NMR Spectroscopy (STD NMR), Mucin peptide-antibody interactions.

Introduction

MUC1 mucin (MUC1 or mucin 1) is a large 1,255 amino acid membrane associated glycoprotein (M_r~200 kDa) expressed on the apical surfaces of various epithelial cells (1,2). The structure of MUC1 is composed of a cleaved shorter transmembrane carboxyl terminal cytoplasmic tail (MUC1-C) that is tethered to a larger N-terminal extracellular domain (MUC1-N) via non-covalent interactions that contains multiple tandem repeat sequences of 20-aa lush with proline, threonine, and serine residues (known as PTS domain) which are O-glycosylated at Ser and/or Thr. MUC1 plays an important role in homeostatic balance on cell surfaces (2,3). MUC1 produced by cancer cells are involved in two other distinct functions 1). in promotion of cellular differentiation, and 2). in signaling mechanisms associated with repression of cellular apoptosis (4,5). Cancer cells, particularly adenocarcinomas, overexpress MUC1 with aberrant glycan chains (6). Variety fragments of the tandem repeat sequence of MUC1 (GVTSAPDTRPAPGSTAPPAH) have been tested as potential cancer vaccines (7,8); several of which are currently in early phases of clinical trials (9,10). Some of these trial vaccine constructs constituted the recognized active epitope PDTRP as the focal sequence in combination with other adjuvants or linker proteins (11,12); others are composed of the glycan antigens and DNA (13,14); yet others are made up of engineered mucin 1 oligopeptides containing multiple valencies (15,16). Multivalent antigens in cancer vaccine field are being sought after.

We examined the binding of the upstream sequence GVTSAPD prior to the active epitope PDTRP containing only one Pro residue to see whether or not such a peptide would still have the ability to bind the mouse Muc1 monoclonal antibody (mAb) (IgG1, 6A4 clone), and if it does, what residue are important for binding and what are the mechanisms for interactions. Once the active part of the peptide is mapped, the objective will be to determine if modification with natural or un-natural side chains can potentially yield new active epitopes that maintain and enhance biological activity. The binding epitopes were mapped by Saturation Transfer Difference NMR (STD NMR) spectroscopy. In a previous study, we showed that the Pro6 residue on GVTSAPD is important for binding to the antibody and substitution of it by Asp (Pro6 replaced by Asp) resulted in a mutant peptide that lost its binding ability. Next, side chains with hydrophobic ring structure were selected for substitution in an effort to gain insight to the structural and physical properties required for binding. The peptides used in this study were GVTSAPD, GVTSADD, GVTSALD, GVTSAYD, GVTSAFD, GVTSAWD and GVTNAPD. We report herein that the epitopes containing hydrophobic ring structure at position-6 of GVTSAPD have the ability to bind mAb.

Abbreviations

HOBt, N-hydroxybenzotriazole; Fmoc, N-α-9-fluorenyl-methoxycarbonyl; Fmoc-Asp(OtBu)-Wang resin (100-200 mesh), N-α-Fmoc-L-aspartic acid β-t-butyl ester-Wang resin; HBTU, O-(Benzotriazol-1-yl)-N,N,N',N'-tetramethyl-uronium hexafluorophosphate; Wang resin, *p*-alkoxybenzyl alcohol-linked Wang resin.

Methods

Reagents and Peptide Synthesis. Fmoc-protected amino acids and reagents (*e.g.*, Fmoc-amino acids, DIPEA, HBTU and HOBt) were purchased from Peptide International and Advanced Chem Tech; pre-loaded resins (Fmoc-Asp(OtBu)-Wang resin), from Novabiochem; solvents, from ACROS Organics/Thermo Fisher Scientific; piperidine, hydrazine, TFA, from Sigma-Aldrich; Kaiser test reagents, from AnaSpec Inc.; NMR solvents, from Cambridge Isotope; Peptides were synthesized manually via the solid-phase technique using standard Fmoc chemistry (17) . For complete details of the synthesis and work-up methods see Yang *et al.* 2013 (18).

Peptide Purification and Mass Spectral Analysis. Peptides were purified via reverse phase HPLC (RAININ HPXL Solvent Delivery System) using a protein and peptide Grace Vydac C_{18} silica-based HPLC column (218TP1010, 10μm, 10x250mm); mass spectra were acquired by an Electro-Spray Ionization Time-of-Flight LC/MS spectrometer (Agilent 6210 Time-of-Flight LC-MS). For complete details of experimental conditions used in HPLC and LC-MS see Yang *et al.* 2013 (18).

NMR Spectroscopy. Procedures for NMR data collection were previously published (18). Only brief account of obligatory

procedures and software are elaborated here. NMR experiments were run with peptide sample concentrations of 2-3 mM in 20 mM phosphate, 5.0 mM NaCl, 90% H_2O, 10% D_2O, pH 5 for protons assignments. A Bruker 400 MHz (9.3977 T) NMR spectrometer with a triple-resonance inverse detection probe and Bruker software TopSpin (v 2.0.5) were used (19). Standard 2D TOCSY and ROESY (mixing time = 300 ms) pulse sequences were employed (20); HDO signal was suppressed by the pulse sequence 3-9-19 WATERGATE (19).

Samples for 1D STD NMR experiments contained 0.10 mM mouse Muc1 IgG, 1.0 mM MUC1 peptide with a peptide:protein ratio of 100:1.0 in 20 mM phosphate buffer, 5.0 mM NaCl, 100 % D_2O, pH 7 at 7°C. The mouse Muc1 specific mAb that recognizes the epitope SAPDTRPA within the core 20-aa Tandem Repeat on MUC1 peptide was used as supplied without further purification. The mAb is an isotype IgG1 (clone 6A4, M_r ~122 kDa) from hybridization of FO myeloma cells with spleen cells from Balb/cj mice (Genway Biotech, Inc.).

STD NMR data were acquired by the 40 Gaussian shaped pulse train for saturation on the f2 channel alternating between on and off resonance (19,21,22). The standard Bruker pulse sequence "stddiff" and another "stddiffesgp.3" with the spoil pulse sequence to destroy unwanted magnetization, water suppression using excitation sculpting with gradients and spinlock (30 ms) to suppress the broad protein background was employed (19,22). The saturation time was 2.00 s with a relaxation delay of 2.10 s. The irradiation power during the presaturation pulse train was 0.0356 mW for all the STD NMR experiments by the probe used. Data were collected as serial pseudo 2D data. The off-resonance FID and on-resonance FID were split, multiplied by the same factor but with a negative factor applied to the on-resonance FID after acquisition, then summed. The resultant difference FID is the STD FID; it was then transformed to a STD NMR spectrum. The STD spectrum (I_{STD}) is obtained by subtracting the off-resonance spectrum (I_0) from the on-resonance spectrum ($I_{on\text{-}res}$), that is $I_{STD} = I_0 - I_{on\text{-}res}$ (21). The on-resonance frequency was set at -2 ppm and the off-resonance frequency at 40 ppm. All STD NMR data were collected with 32 dummy scans, 1024 scans by $512t_1$ points with a spectral window of 5028.3 Hz. Control STD NMR spectra were carried out with the corresponding peptide at the same concentration using the same pulse sequence and parameters without the antibody to make sure that there was no spurious saturation transfer peaks due to excessive saturation power. The same conditions and spectral parameters were used on the peptide-antibody mixture as those used for the control peptide that resulted in a flat STD NMR spectrum. The amplification factor (A_{STD}) is calculated by the equation, $A_{STD} = (I_{STD}/I_0) \times ([L]_t/[P])$, where I_0 is the intensity of the control

Table 1. ^{1}H chemical shifts (ppm) assigned for all the mucin peptides used in the experiments, acquired in 20 mM phosphate, 5 mM NaCl, 90% H_2O, 10% D_2O, pH 5.0, at 7 °C in the absence of MUC1 mucin monoclonal antibody (6A4) referenced to the HDO resonance at 4.97 ppm at 7 °C.

Residue	NH	αH	βH	Others
GVTSAPD				
G	-	3.887		
V	8.644	4.253	2.118	γ, γ'CH_3 0.955, 0.938
T	8.491	4.422	4.231	γCH_3 1.209
S	8.464	4.439	3.842	
A	8.576	4.614	1.367, 2.715	
P		4.422	2.257, 1.993	Hγ 1.993; Hδ,δ' 3.657, 3.790
D	8.119	4.366	2.619, 2.715	
GVTNAPD				
G	8.068	3.895		
V	8.627	4.235	2.095	γ, γ'CH_3 0.942, 0.942
T	8.483	4.352	4.162	γCH_3 1.187
N	8.627	4.697	2.689, 2.771	
A	8.382	4.569	1.352	
P		4.416	2.281, 1.966	Hγ 1.992; Hδ,δ' 3.653, 3.778
D	8.452	4.634	2.902, 2.902	
GVTSAWD				
G	-	3.859		
V	8.623	4.185	2.080	γ, γ'CH_3 0.912, 0.912
T	8.497	4.364	4.185	γCH_3 1.142
S	8.518	4.364	3.665	
A	8.352	4.185	1.142	
W	7.966	4.688	3.324, 3.171	Hδ 6.939; Hε 7.381; Hζ,ζ' 6.872, 7.199; Hη 6.939
D	7.786	4.356	2.625, 2.625	
GVTSAFD				
G	-	3.866		
V	8.640	4.241	2.113	γ, γ'CH_3 0.9393, 0.9567
T	8.515	4.387	4.204	γCH_3 1.216
S	8.448	4.410	3.785, 3.841	
A	8.434	4.226	1.230	
F	8.201	4.625	2.955, 3.199	Hδ,δ' 7.250; Hε,ε' 7.343; Hζ,ζ' 7.297
D	8.053	4.507	2.795, 2.811	
GVTSADD				
G	-	3.881		
V	8.362	4.247	2.120	γ, γ'CH_3 0.956, 0.939
T	8.487	4.330	4.247	γCH_3 1.211
S	8.487	4.330	3.886, 3.844	
A	8.487	4.325	1.387	
D	8.402	4.702	2.886, 2.746	
D	8.052	4.563	2.839, 2.839	
GVTSAYD				
G	-	3.847		
V	8.642	4.215	2.097	γ, γ'CH_3 0.928, 0.928
T	8.503	4.390	4.208	γCH_3 1.176
S	8.452	4.416	3.798	
A	8.448	4.201	1.223	
Y	8.137	4.571	3.126, 2.857	Hδ,δ' 7.114; Hε,ε' 6.810
D	7.958	4.443	2.743, 2.743	
GVTSALD				
G	-	3.895		
V	8.659	4.249	2.135	γ, γ'CH_3 0.972, 0.972
T	8.502	4.440	4.249	γCH_3 1.229
S	8.484	4.480	3.659, 3.721	
A	8.535	4.339	1.417	
L	8.344	4.367	1.638	γ, γ'CH_3 0.872, 0.932
D	7.914	4.385	2.542, 2.645	

peak without saturation, I_{STD}, the peak intensity of the STD NMR peak, and $[L]_t/[P]$, the ratio of the total ligand concentration to antibody concentration (21,22).

Results

Peptide Synthesis, HPLC Purification and Mass Spectral Analysis. Crude peptide yield, in general, were >74% (from HPLC peak integration) with one main peptide peak isolated from impurities. The peptides used for these experiments were HPLC purified to a single peak with >90-95% purity (HPLC peak integration). The $[M+H]^+$ ions of the purified peptides are: for GVTSAPD, 646.540 u versus 646.667 u (theoretical mass); for GVTSAWD, 735.351 u versus 735.764 u (theoretical mass); for GVTSAFD, 696.336 u versus 696.727 u (theoretical mass); for GVTSADD, 664.285 u versus 664.639 u (theoretical mass); for GVTSALD, 662.384 u versus 662.710 u (theoretical mass); for GVTNAPD, 673.315 u versus 673.693 u (theoretical mass); and for GVTSAYD, 712.314 u versus 712.726 u (theoretical mass).

NMR Proton Assignments. 2D TOCSY ^{1}H NMR data were used for spin system assignments and ROESY data, for detection of interresidue proton-proton interactions (e.g., $H\alpha_{(i)} \rightarrow NH_{(i+1)}$, $d_{H-H} \leq 5Å$). Proton assignments for all peptides were completed in entirety and unambiguously, except for stereospecificity assignments. Table 1 shows the chemical shift (ppm) values assigned to all the protons of each peptide at 90% H_2O, 10% D_2O. The chemical shifts of aliphatic protons of each peptide obtained under the noted condition (pH 5.0, 90% H_2O, 10% D_2O, 7°C) did not deviate indistinguishably from those obtained in pH 7.0, 100% D_2O, 7°C in the presence of MUC1 mAb. Thus, protons having saturation transfer effect (on STD NMR spectra) were readily recognizable from these assignments.

STD NMR Results. The STD NMR data showed that the native mucin peptide GVTSAPD, though truncated and not containing the full binding epitope PDTRP, still retains interactions with the Muc1 mAb (6A4). The binding results of GVTSAPD and of a non-binding peptide GVTSADD were previously published by the corresponding investigator of this work (18). That work showed that Pro6 residue in GVTSAPD is important for binding. Figure 1 shows the STD NMR results obtained for the peptides under this study.

The STD NMR results showed that saturation transfer effects occurred at the side chain protons of the Pro6 residues of GVTSAPD and GVTNAPD (Figure. 1, panel A, traces A and C). Table 2 lists estimates of the peak intensities due to saturation transfer effect as determined by the amplification factor (A_{STD}).

Table 2: Amplification factors (A_{STD}) estimated for each proton on the amino acid residues of each peptide that showed observable STD peak; the dash '-' represents no observation of STD peak and taken to be zero intensity.

Residues Ha	Hα	Hβ	Hγ	Hδ	Sum of A_{STD} per each residue
GVTSAPD					
G	1.4[a]				1.4
V	ne	-	0.9		0.9
T	3.1[b]	ne	1.3		4.4
S	3.1[b]	0.8[c]			3.9
A	1.8	2.4			4.2
P	3.1[b]	5.4	9.9	ne	18.4
D	3.3	2.2			5.5
ne, not estimated due to overlapping; [a] overlapped between $G_1H\alpha$ and $S_4H\beta$; [b] overlapped between Hα's of T_3, S_4 and P_6; [c] overlapped between $S_4H\beta$ and $P_6H\delta$.					
GVTNAPD					
G	0.4				0.4
V	0.1	-	1.1		1.1
T	0.7	0.2	1.1		2.0
N	1.4	-			1.4
A	2.0[a]	2.4			4.4
P	5.2	7.9	11.9	2.1	27.1
D	2.0[a]	5.9			7.9
[a] overlapped between $A_5H\alpha$ and $D_7H\alpha$.					
GVTSAWD					
G	-				0
V	-	-	0.1		0.1
T	-	-	0.3		0.3
S	-	-			
A	-	2.3			2.3
W	-	-	19.5[a]		19.5
D	-	-			0
[a] sum of A_{STD} of all aromatic protons on the indole ring.					
GVTSAFD					
G	-				0
V	-	-	-		0
T	-	-	-		0
S	-	-			0
A	-	-			0
F	-	-	10.5[a]		10.5
D	-	-			0
[a] sum of A_{STD} of all aromatic protons on the phenyl ring.					
GVTSAYD					
G	0.1				0.1
V	0.1[a]	-	1.9		1.9
T	-	0.1[a]	1.7[b]		1.8
S	-	-			0
A	0.1[a]	1.7[b]			1.8
Y	-	-	1.0[c]		1.0
D	-	-			0
[a] overlapped between $V_2H\alpha$, $A_5H\alpha$ and $T_3H\beta$; [b] overlapped between methyl groups of T_3 and A_5; [c] sum of A_{STD} of all aromatic protons on the phenol ring.					

Similar STD effects were obtained for the peptides GVTNAPD and GVTSAPD. The STD peak intensity of Pro6 side chain protons centered at the Hγ's and one of the Hβ's was strongest, having A_{STD} values of 11.9 and 9.9 (Table 2). The A_{STD} values of Hβ's were lower, being 7.9 and 5.4, those of the Hα's were 5.2 and 3.1, and essentially none at Hδ's. Due to resonances overlapping at Hγ, Hα and Hβ, it is not possible to explicitly estimate the A_{STD} of these single protons of Pro6,

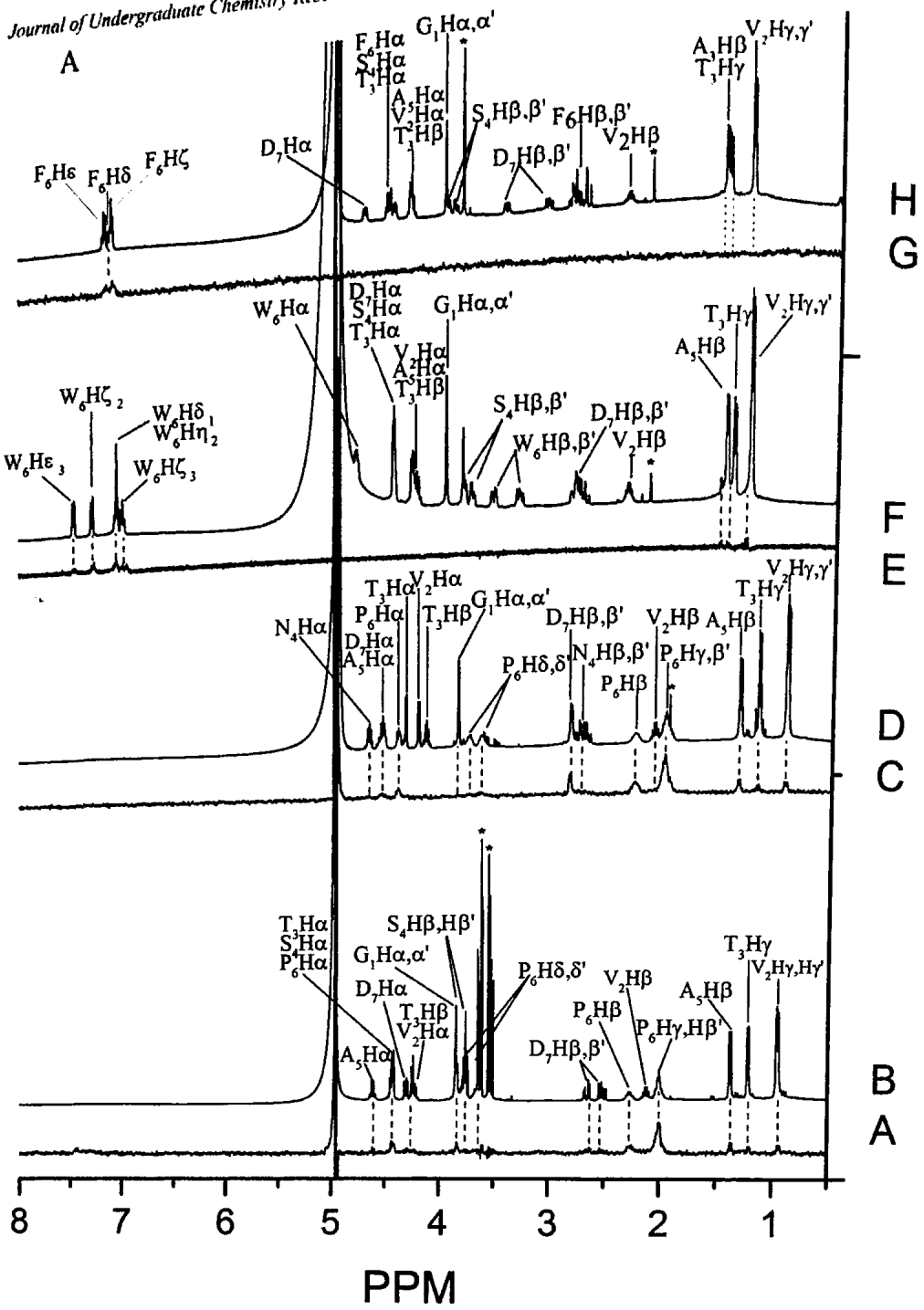

Figure 1. Panel A. Pairs of ¹H STD NMR spectra (bottom) and reference ¹H NMR spectra (top) in the presence of mAb (6A4) in 20 mM phosphate, 5 mM NaCl, pH 7.0, 100% D_2O, 7°C. Dashed lines connect the protons that experienced STD effect on the STD NMR spectra to the corresponding protons on the ¹H NMR spectra. The pairs of spectral traces are: A&B). for GVTSAPD peptide, with highest STD peak intensity located at Pro6Hδ resonance, its intensity was adjusted to the same height as the one in the normal ¹H NMR spectrum; note that contaminants at 3.6 ppm were subtracted out in the STD spectra as they do not interact with the mAb; C&D). for GVTNAPD peptide, with similar STD effects to spectrum (A); E&F). for GVTSAWD peptide; G&H). for GVTSAFD peptide. Contaminants are marked by asterisks (*) on both panels.

and this is true for other overlapping as well. Nonetheless, it is clear that the peak intensities are higher where the protons of Pro6 are located on the spectrum (except Hδ's), for instance, an isolated single Hβ of Val2 (at 2.12 ppm) located between the Hβ and Hγ of Pro6 did not experience any saturation transfer effect, so it had no STD peak. Weak saturation transfer effects were also detected at the $-CH_3$'s of Val2, Thr3 and Ala5, and at Hα of Ala5, with noted slightly higher peak intensity at Hβ's of Asp7 on GVTNAPD. In both of these peptides, the total A_{STD} per residue is highest for the Pro6 residue and dropped off on either side of it (Table 2).

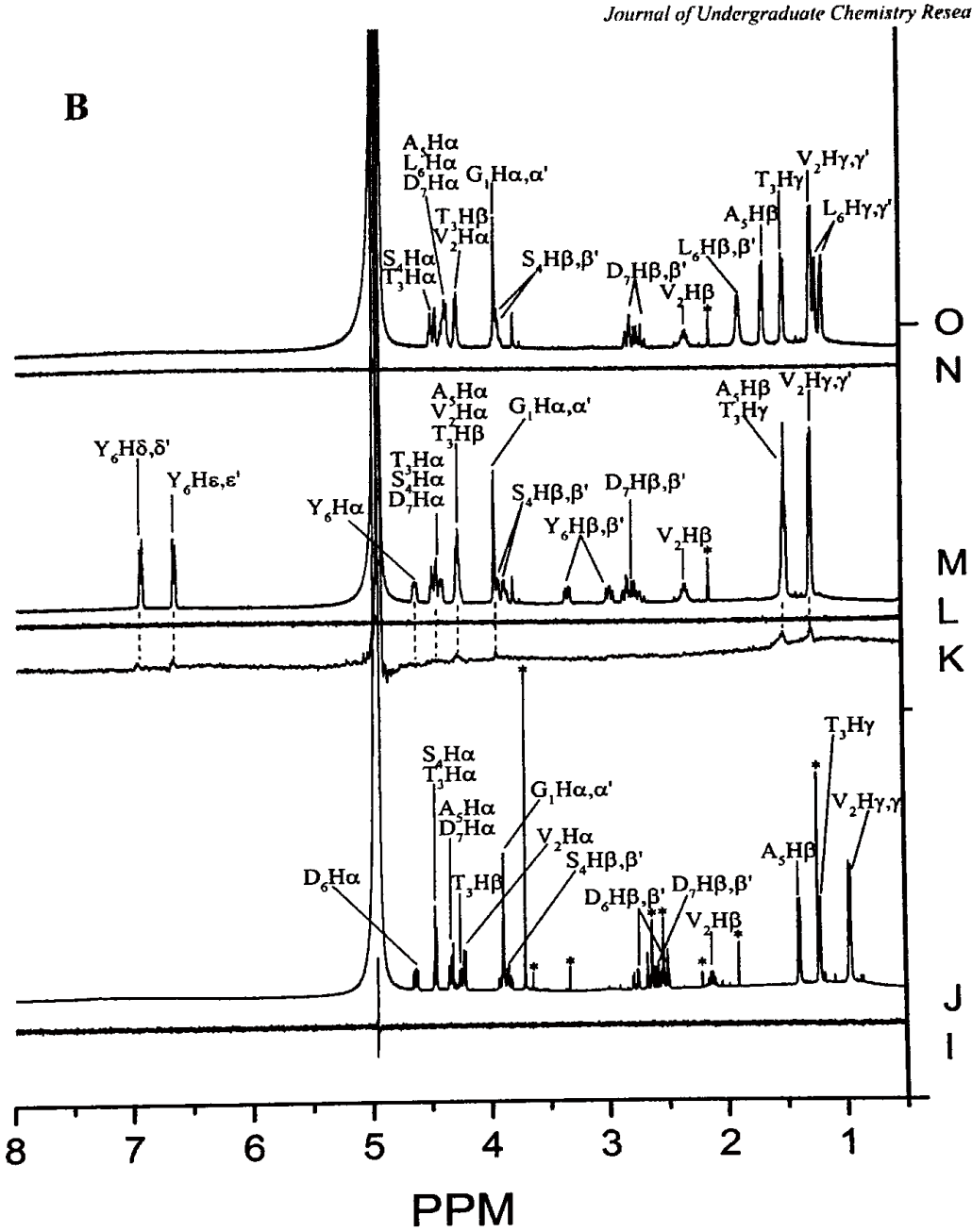

Figure 1. Panel B. Pairs of ^{1}H STD NMR spectra (bottom) and reference ^{1}H NMR spectra (top) in the presence of mAb. The pairs of spectral traces are: I&J). for GVTSADD peptide, showing flat line for STD NMR spectrum; K, L&M). for GVTSAYD peptide; (K), same spectrum as L but without protein background subtraction; N&O). for GVTSALD.

The peptides in which Pro6 is substituted by Phe and Trp showed markedly different STD NMR results. Some saturation transfer effects were experienced by the aromatic protons at Hδ, Hε and Hζ of the phenyl ring of Phe6 of GVTSAFD (A_{STD} = 10.5), and also by the aromatic protons Hδ, Hε, Hζ's and Hη of the indole ring of Trp6 of GVTSAWD (A_{STD} = 19.5) (Fig. 1, Panel A, traces E and G). Only very weak peak intensities were observed at the methyl groups of Val2, Thr3 and Ala5. When Pro6 was replaced by Tyr very weak STD peak intensities were detectable at the methyl and aromatic protons (Fig. 1, Panel B, Trace K). The protons of two peptides in which Pro6 was substituted by Asp or Leu experienced no saturation transfer effects to any of them (Fig. 1, Panel B, traces I and N).

Discussion

Based to the STD peak intensity, which signifies that the higher the peak intensity the closer that proton is to the binding site or otherwise (21,22), the interactions of Pro6 side chain protons of GVTSAPD and GVTNAPD placed the Hγ closest to the antibody binding interface followed by Hβ and Hα. Substitution of a polar residue Ser4 with another polar residue Asn, similar in side chain size and physical properties while

leaving the Pro6 alone, does not affect the binding ability of that peptide with the antibody. It was not practical to determine the K_d value due to limited supply of the mAb samples employing the STD NMR method. Nonetheless, the binding of these peptides are likely to fall in the weak binding range as determined for related mucin peptides previously reported by other workers [K_d ~ 0.063-11 μM] (23,24). The side chain protons of other residues (Val2, Thr3, Ser4, Ala5 and Asp7) that showed weak STD peak intensities would be situated further away from the binding interface. For the peptides GVTSAWD and GVTSAFD, mainly the aromatic protons experienced saturation transfer effects, which indicated that those protons were closer in the vicinity of the binding interface, that is the aromatic rings were in contacts at the binding site.

On the peptide in which Pro6 is substituted by Tyr, the aromatic protons received no saturation transfer effect, however in a separate STD NMR data without protein background subtraction minute saturation transfer effects were visible at the aromatic and methyl regions (Figure 1, Panel B, spectrum K). This signifies that perhaps the peptide had very weak interaction with the antibody. The peptides with Leu and Asp substituted for Pro6 were considered non-binders since their protons did not experience any saturation transfer effects. Even though, the side chain of Leu is hydrophobic, its isobutyl structure may not give the correct structural properties (e.g., not being a cyclic structure) necessary for interacting with the binding residues existing at the binding interface. Similar argument can be drawn for the Asp residue for being polar and non-cyclic. Using the crystal structure of MUC1 monoclonal antibody SM3 complexed with the 13-aa mucin peptide (1-TSAPDTRPAPGST-13) as a peptide-antibody binding model, the Pro4 side chain atoms are locked in by two Trp and one Tyr residues at the binding pocket (25). It is not known what residues line the antigen binding pocket in the mAb 6A4; if its environment at the binding interface is similar to that of the SM3, then this provides an explanation to support position-6 for favoring cyclic hydrophobic residue for binding, which helps to explain the binding properties that we observed for the peptides under investigation. The results from this study indicate that at the Pro6 position if a group containing hydrophobic ring structure is substituted for Pro, the resultant peptide will have the ability to bind the antibody. Thus, hydrophobic properties and ring structure at the Pro6 position appear to play key role for binding.

Conclusions

The results from this study revealed that Pro-substituted peptides GVTSAFD and GVTSAWD have the ability to bind mAb (6A4), but not GVTSADD and GVTSALD. The shortened mucin peptide GVTSAPD near the main binding epitope, despite lacking most of the binding residues, can still bind mAb; its residue Pro6 is required for binding. When a less critical residue Ser4 is substituted by Asn (GVTNAPD), no change in the binding effect occurred. Ring structure with hydrophobic properties is important for binding at Pro6; substitution of it by Phe or Trp resulted in binding but substitution by Asp or Leu showed otherwise. The peptide GVTSAYD showed minute STD peak intensities at the aromatic and methyl regions compare to those of GVTSAFD. To date, the development of mucin-based antigenic agents (e.g., vaccines) to induce effective immunity against certain type of cancers (e.g., adenocarcinomas) has yet to yield such a product, though some are being tested in clinical trials (9). Multivalency MUC1 antigens are being sought in the development of cancer vaccine. Substituted mucin peptides such as GVTSAFD and GVTSAWD and their derivatives (cyclic mucin peptides) that have binding ability to MUC1 antibody may provide useful additional epitopes toward designing such agents.

Conflict of Interest

Each author of this manuscript declares he/she has none whatsoever conflict of interest on the materials being printed.

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